Armon × Gauthier × Rick × Julien

1 To do

- check tradable/non tradable in % of EU consumption
- Labor income tax, export rebate?
- Export/import EoS for commodities?
- kill the trends of inflation target...
- Targeted subsidies?
- What is home and foreign ? EU vs US ? EU vs small open econ ? EU vs RoW ?
- Discuss firm entry !!

2 Literature

- Sahuc et al. (2024): Integrates climate into a New Keynesian (NK) framework. Disentangles *greenflation* from *climateflation* and analyzes the optimal monetary response to climate shocks.
- Bergin and Corsetti (2023): Demonstrates that tariff shocks differ from traditional supply shocks. Using a stylized supply chain dimension, they show that monetary expansion is the optimal response to offset the distortionary nature of tariffs.
- Bergin and Corsetti (2025a): Analyzes a two-sector model (differentiated/sticky prices vs. non-differentiated/flexible prices). Highlights that optimal monetary policy requires coordination, as price reactions vary significantly across sectors.
- Bergin and Corsetti (2025b): Explores how monetary policy favors reallocation induced by tariffs. Expansionary policy triggers depreciation to offset relative price increases. Emphasizes the role of comparative advantage and currency dominance in a tariff war scenario.
- Ambec et al. (2024): Examines welfare effects and efficiency of CBAM in an industry producing homogeneous polluting goods. Shows that export rebates and marginal cost heterogeneity determine whether a no-trade equilibrium emerges.
- Clausing et al. (2025): Quantifies CBAM effects (focusing on steel/aluminum) using trade and production elasticities. CBAM shifts the marginal abatement cost (MAC) and provides incentives for foreign regulatory adaptation.
- Coster et al. (2024): Develops an input sourcing model to estimate carbon leakage. Includes a disutility term for climate damages. Highlights the impact of CBAM on firm-level sourcing decisions under monopolistic competition.

3 Research question

What are the effects of CBAM ?

Is CBAM inflationary? How production is relocated? Is there some substitution from intermediate importation to final good importation? What is the effect on total emissions?
How do all of these effects vary with carbon tax levels and to crucial parameter (technology, abatement costs, elasticities...).

One key mechanism: foreign firms have two strategies facing CBAM: i) decrease their price by reducing their markup ii) abate more. It will be a function of demand elasticity, marginal cost of abatement, share of exportation of the firm.

How to answer the questions: simulations of the model with and without CBAM to observe how the economy reacts, compare the reaction of all macro variables.

What is the optimal monetary policy response?

Once we accurately asses the effect of CBAM on the economy we can compare different monetary policy regimes in terms of welfare for both countries, the benchmark is a simple Taylor rule response in both countries.
We can compare it to:
i) Unilateral welfare maximization of a country : given the other country Taylor rule, what is the optimal monetary policy.
ii) Ramsey cooperative monetary policy, maximize the sum of Welfare in both countries.
iii) More complicated: strategic response from both countries that want to maximize their welfare given the other's reaction, in the spirit of a Cournot duopoly?

How to answer the questions: we can look for analytical solution of optimal policy if possible otherwise compute it numerically by maximizing welfare. Compare what are the monetary policies responses in theses cases and compare welfare.

What is the optimal carbon tax response from foreign country?

Facing CBAM, the foreign country has the ability to adjust its carbon tax, what is the optimal response?
When is there some incentive to increase $\tau_e(f)$? Given the fact that the foreign firm will suffer from the CBAM, the foreign country needs to decide whether increasing the tax is more or less costly. It will affect of consumers in its countries but at least the carbon taxe revenues will be redistributed lump-sump in its country.

How to answer the question: Solve numerically the model, compute welfare in each country. Under what condition there is incentive for an increase in $\tau_e(f)$? Are $\tau_e(h)$ and $\tau_e(f)$ complements or substitutes?
Heat map of welfare as a function of both carbon taxes.

Link with "capturing carbon rents"

4 General questions

- CBAM applies only to specific products; how does the nature of these products (intermediates/final, differentiated/non differentiated) change the transmission?
- Does CBAM trigger technological progress? What effect on abatement decisions?
- Should we/How explicitly model the reorganization of supply chains triggered by carbon pricing?
- Bergin & Corsetti emphasize cooperative Ramsey policies. Does it apply to CBAM? Interesting as a point of comparison but I can imagine retaliation or unilateral policy to offset CBAM.
- Monetary policy tries to offset tariff distorsions through exchange rates. CBAM actually want to decrease trade of carbon intensive products, how this changes the optimal policy?
- How are CBAM revenues used?
- How CBAM path affects price-setting by firms?
- Do we need a damage function? Country-specific damages? Shouldn't the damage affect both final goods and intermediate goods production?

5 Model

5.1 Households

Households maximize lifetime utility subject to employment status $\theta \in \{0, 1\}$:

$$\max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left\{ \frac{c_{i,t}^{1-\sigma}}{1-\sigma} - \psi_t \frac{n_{i,t}^{1+\sigma_n}}{1+\sigma_n} \right\}$$

The consumption aggregate:

$$c_{i,t} = \left((1-\gamma_c)^{\frac{1}{\phi}} c_{i,t}(h)^{\frac{\phi-1}{\phi}} + \gamma_c^{\frac{1}{\phi}} c_{i,t}(f)^{\frac{\phi-1}{\phi}} \right)^{\frac{\phi}{\phi-1}}$$

Price index:

$$P_t = \left[(1-\gamma_c) p_t(h)^{1-\phi} + \gamma_c (e_t p_t(f))^{1-\phi} \right]^{\frac{1}{1-\phi}}$$

Budget Constraints:

$$\begin{aligned} \theta_{i,t} = 1: \quad & c_{i,t} + b_{i,t}(h) + e_t b_{i,t}(f) + T_{i,t}^S(h) = \frac{r_{t-1}(h)}{\pi_t} b_{i,t-1}(h) + \\ & e_t \frac{r_{t-1}(f)}{\pi_t} b_{i,t-1}(f) + AC(b_{i,t}(f)) + \Pi_{i,t} + w_t n_{i,t} + T_{i,t}^e \\ \theta_{i,t} = 0: \quad & c_{i,t} + b_{i,t}(h) + e_t b_{i,t}(f) = \frac{r_{t-1}(h)}{\pi_t} b_{i,t-1}(h) + \\ & e_t \frac{r_{t-1}(f)}{\pi_t} b_{i,t-1}(f) + AC(b_{i,t}(f)) + d_{i,t}(h) + T_{i,t}^e \end{aligned}$$

5.1.1 Solve: Household Decisions

1. Home Households
Intratemporal decision: The household minimizes the cost of the consumption basket.

$$\min_{c_{i,t}(h),c_{i,t}(f)} \quad p_t(h)c_{i,t}(h) + e_t p_t(f)c_{i,t}(f)$$
$$\text{s.t.} \quad c_{i,t} = \left((1-\gamma_c)^{\frac{1}{\phi}} c_{i,t}(h)^{\frac{\phi-1}{\phi}} + \gamma_c^{\frac{1}{\phi}} c_{i,t}(f)^{\frac{\phi-1}{\phi}}\right)^{\frac{\phi}{\phi-1}}$$

Setting up the Lagrangian yields the optimal demand functions for domestic and imported goods:

$$c_{it}(h) = (1-\gamma_c)\left(\frac{P_t}{p_t(h)}\right)^{\phi} c_{it} \quad \text{and} \quad c_{it}(f) = \gamma_c\left(\frac{P_t}{e_t p_t(f)}\right)^{\phi} c_{it}$$

where the associated Consumer Price Index (CPI) is:

$$P_t = \left[(1-\gamma_c)p_t(h)^{1-\phi} + \gamma_c(e_t p_t(f))^{1-\phi}\right]^{\frac{1}{1-\phi}}$$

Intertemporal decision: The high productivity household ($\theta_{i,t} = 1$) maximizes expected lifetime utility.

$$\begin{aligned}
\mathcal{L} = \mathbb{E}_0 &\left\{\sum_{t=0}^{\infty} \beta^t \left(\frac{c_{i,t}^{1-\sigma}}{1-\sigma} - \psi_t \frac{n_{i,t}^{1+\sigma_n}}{1+\sigma_n}\right)\right\} \\
&+ \mathbb{E}_0 \sum_{t=0}^{\infty} \lambda_{i,t}\left[\frac{r_{t-1}(h)}{\pi_t} b_{i,t-1}(h) + e_t \frac{r_{t-1}(f)}{\pi_t} b_{i,t-1}(f) + \Pi_{i,t} + w_t n_{i,t} + T_{i,t}^e \right. \\
&\left. - \left(c_{i,t} + b_{i,t}(h) + e_t b_{i,t}(f) + T_{i,t}^S(h) + AC(b_{i,t}(f))\right)\right]
\end{aligned}$$

Taking derivatives with respect to $c_{i,t}$, $n_{i,t}$, $b_{i,t}(h)$, and $b_{i,t}(f)$ yields the following system of first-order conditions:

$$\begin{cases}
c_{i,t}^{-\sigma} = \beta \mathbb{E}_t\left[c_{i,t+1}^{-\sigma} \dfrac{r_t(h)}{\pi_{t+1}}\right] \\
\psi_t n_{i,t}^{\sigma_n} = c_{i,t}^{-\sigma} w_t \\
\dfrac{r_t(h)}{\pi_{t+1}} = e_{t+1}\dfrac{r_t(f)}{\pi_{t+1}} + \dfrac{\partial AC(b_{i,t}(f))}{\partial b_{i,t}(f)}
\end{cases}$$

Euler equation for the employed type ($q \in (H, L)$) is:

$$c_{i,q,t}^{-\sigma} = \mathbb{E}_t\left\{\beta \frac{r_t(h)}{\pi_{t+1}}\left((1-\omega)c_{i,H,t+1}^{-\sigma} + \omega c_{i,L,t+1}^{-\sigma}\right)\right\}$$

Interpretation:

- *Euler equation* Marginal utility of current consumption = discounted expected utility of future consumption.
- *Labor supply:* Marginal disutility of labor = marginal utility of labor income.
- *Bonds:* Domestic returns = Foreign real return net of holding foreign bonds costs.

2. Foreign Households
Foreign households face a symmetric problem, denoting foreign variables with an asterisk (*). The price of imported home goods in foreign currency is $p_t(h)/e_t$.
By symmetry, the optimal demands and price index are:

$$c_{i,t}^*(f) = (1-\gamma_c^*)\left(\frac{P_t^*}{p_t^*(f)}\right)^{\phi} c_{i,t}^*, \quad c_{i,t}^*(h) = \gamma_c^*\left(\frac{P_t^*}{p_t(h)/e_t}\right)^{\phi} c_{i,t}^*$$

$$P_t^* = \left[(1-\gamma_c^*)p_t^*(f)^{1-\phi} + \gamma_c^*\left(\frac{p_t(h)}{e_t}\right)^{1-\phi}\right]^{\frac{1}{1-\phi}}$$

Assuming foreign households face an adjustment cost $AC(b_{i,t}^*(h))$ when holding home bonds, FOC are:

$$\begin{cases} {c_{i,t}^*}^{-\sigma} = \beta\mathbb{E}_t\left[{c_{i,t+1}^*}^{-\sigma}\frac{r_t(f)}{\pi_{t+1}^*}\right] \\ \psi_t^*(n_{i,t}^*)^{\sigma_n} = (c_{i,t}^*)^{-\sigma}w_t^* \\ \frac{r_t^*(f)}{\pi_{t+1}^*} = \frac{1}{e_{t+1}}\frac{r_t(h)}{\pi_{t+1}^*} + \frac{\partial AC(b_{i,t}^*(h))}{\partial b_{i,t}^*(h)} \end{cases}$$

Euler equation for the employed type ($q \in (H,L)$) is:

$$(c_{i,q,t}^*)^{-\sigma} = \mathbb{E}_t\left\{\beta\frac{r_t^*(f)}{\pi_{t+1}^*}\left((1-\omega)(c_{i,H,t+1}^*)^{-\sigma} + \omega(c_{i,L,t+1}^*)^{-\sigma}\right)\right\}$$

5.2 Final Good Sector

Two countries, home (h) and foreign (f). Production uses domestic and imported intermediates:

$$y_t(x) = \alpha_x[G_t(x)]^{\zeta} l_t^{1-\zeta}$$

where $G_t(x)$ is a CES aggregator with substitutability η:

$$G_t(h) = \left((1-\gamma_y)^{\frac{1}{\eta}}\int_0^{l_t(h)} y_{j,t}(h)^{\frac{\eta-1}{\eta}}dj + \gamma_y^{\frac{1}{\eta}}\int_0^{l_t(f)} y_{j,t}(f)^{\frac{\eta-1}{\eta}}dj\right)^{\frac{\eta}{\eta-1}}$$

$$G_t(f) = \left({\gamma_y^*}^{\frac{1}{\eta}}\int_0^{l_t(h)} y_{j,t}(h)^{\frac{\eta-1}{\eta}}dj + (1-\gamma_y^*)^{\frac{1}{\eta}}\int_0^{l_t(f)} y_{j,t}(f)^{\frac{\eta-1}{\eta}}dj\right)^{\frac{\eta}{\eta-1}}$$

The profit of the final good firm is, taking into account iceberg cost of trade (τ_i):

$$\begin{aligned}\Pi_t(h) = {} & p_t(h)\alpha_h l_t^{1-\zeta}\left[G_t(h)\right]^{\zeta} - \int_0^{l_t(h)} p_{j,t}(h)y_{j,t}(h)dj \\ & - e_t\int_0^{l_t(f)}(1+\tau_i)p_{j,t}(f)y_{j,t}(f)dj \\ & - e_t\int_0^{l_t(f)}\max\{\tau_{e,t}(h)-\tau_{e,t}(f),0\}\,\sigma_t(f)(1-\mu_{j,t}(f))y_{j,t}(f)dj\end{aligned}$$

CBAM enters profit through a carbon tax on top of foreign one, meaning on $p_{i,j,t}(f)$.

5.2.1 Solve

1st step: choose between foreign and domestic intermediates.
For home firms, let the effective price of foreign intermediates be $\tilde{p}_{j,t}(f)$:

$$\tilde{p}_{j,t}(f) \equiv (1+\tau_i)p_{j,t}(f) + \max\{\tau_{e,t}(h) - \tau_{e,t}(f), 0\}\,\sigma_t(f)(1-\mu_{j,t}(f))$$

The cost minimization problem is:

$$\min_{y_{j,t}(h),y_{j,t}(f)} \quad \int_0^{l_t(h)} p_{j,t}(h)y_{j,t}(h)dj + e_t\int_0^{l_t(f)} \tilde{p}_{j,t}(f)y_{j,t}(f)dj$$

$$\text{s.t.} \quad G_t(h) = \left((1-\gamma_y)^{\frac{1}{\eta}}\int_0^{l_t(h)} y_{j,t}(h)^{\frac{\eta-1}{\eta}}dj + \gamma_y^{\frac{1}{\eta}}\int_0^{l_t(f)} y_{j,t}(f)^{\frac{\eta-1}{\eta}}dj\right)^{\frac{\eta}{\eta-1}}$$

Lagrangian:

$$\begin{aligned}\mathcal{L} = &\int_0^{l_t(h)} p_{j,t}(h)y_{j,t}(h)dj + \int_0^{l_t(f)} e_t\tilde{p}_{j,t}(f)y_{j,t}(f)dj \\ &+ P_{G,t}\left[G_t(h) - \left((1-\gamma_y)^{\frac{1}{\eta}}\int_0^{l_t(h)} y_{j,t}(h)^{\frac{\eta-1}{\eta}}dj + \gamma_y^{\frac{1}{\eta}}\int_0^{l_t(f)} y_{j,t}(f)^{\frac{\eta-1}{\eta}}dj\right)^{\frac{\eta}{\eta-1}}\right]\end{aligned}$$

First-Order Conditions:

$$\begin{aligned}\frac{\partial\mathcal{L}}{\partial y_{j,t}(h)} = 0 \quad &\Longrightarrow \quad y_{j,t}(h) = (1-\gamma_y)\left(\frac{p_{j,t}(h)}{P_{G,t}}\right)^{-\eta} G_t(h) \\ \frac{\partial\mathcal{L}}{\partial y_{j,t}(f)} = 0 \quad &\Longrightarrow \quad y_{j,t}(f) = \gamma_y\left(\frac{e_t\tilde{p}_{j,t}(f)}{P_{G,t}}\right)^{-\eta} G_t(h)\end{aligned}$$

which yields the price index:

$$P_{G,t} = \left((1-\gamma_y)\int_0^{l_t(h)} p_{j,t}(h)^{1-\eta}dj + \gamma_y\int_0^{l_t(f)} \left(e_t\tilde{p}_{j,t}(f)\right)^{1-\eta}dj\right)^{\frac{1}{1-\eta}}$$

Interpretation:

- *Demand for intermediates:* variety j = determined by its price relative to the aggregator $P_{G,t}$, with elasticity η governing the substitutability between domestic and foreign inputs.
- *Effective import price:* $\tilde{p}_{j,t}(f)$ = total cost of foreign inputs including trade frictions (iceberg costs) and the CBAM.

2nd step: maximise the profit of the final good firm

$$\max_{G_t(h)} \quad \Pi_t(h) = p_t(h)\alpha_h l_t^{1-\zeta} G_t(h)^{\zeta} - P_{G,t}G_t(h)$$

Which yields the FOC:

$$G_t(h) = \left(\frac{\zeta p_t(h)\alpha_h l_t^{1-\zeta}}{P_{G,t}}\right)^{\frac{1}{1-\zeta}}$$

Foreign final good firms optimality equations are analogous.
Interpretation:

- *Demand for intermediates* marginal cost ($P_{G,t}$) equals the marginal value product.

5.3 Intermediate Good Sector

Intermediate goods are produced by monopolistic firms:

$$y_{j,t}(x) = \Gamma_t(x) l_{j,t}(x)^{1-\alpha} (n_{j,t}(x)^d)^\alpha$$

where the TFP term $\Gamma_t(x) = z_t(x)\Phi_x(m_t)$ includes the damage function $\Phi_x(m_t) = \exp(-\gamma_x(m_t - m_{1750}))$. Emissions follow:

$$e_{j,t}(x) = \sigma_t(x)(1 - \mu_{j,t}(x))y_{j,t}(x)$$

The country-specific abatement cost is:

$$C^a_{j,t}(x) = \theta_{1,t}(x)\mu_{j,t}^{\theta_2(x)} y_{j,t}(x) \quad \text{where} \quad \theta_{1,t}(x) = \frac{p_b(x)}{\theta_2(x)}(1 - \delta_{pb}(x))^{t-t_0}\sigma_t(x)$$

With a country specific emission level, carbon taxation and abatement costs it is possible how exports and imports are affected by different levels of carbon taxation with and without carbon adjustment mechanism.
We can calibrate emissions and costs to match worldwide differences in marginal costs and emission-intensity in production processes.

Profit of intermediates firms is such :

$$\max \pi_{j,t}(x) = p_{j,t}(x)y_{j,t}(x) - w_t(x)\left(\frac{y_{j,t}(x)}{\Gamma_t(x)}\right)^{1/\alpha} - \underbrace{C^a_{j,t}(x)}_{\text{abatement}} - \underbrace{\tau_{e,t}(x)\sigma_t(x)(1 - \mu_{j,t}(x))y_{j,t}(x)}_{\text{carbon tax}}$$

Firms face Rotemberg adjustment costs:

$$C^p_{j,t}(x) = \frac{\kappa}{2}\left(\frac{p_{j,t}(x)}{p_{j,t-1}(x)} - \pi^\star\right)^2 \frac{y_t(x)}{l_t(x)}$$

Death shock (exogenous exit of firm, Bilbiie et al. (2012); Del Negro et al. (2023)) with probability $\vartheta \in (0, 1)$. $\omega_{j,t} = \{0, 1\}$ indicates if the firm exited the market or not. Intermediate goods firms producer maximize:

$$\max_{\{p_{j,t}\}} \mathbb{E}_t \left\{ \sum_{s=0}^{\infty} \beta^s \omega_{j,t+s} \left(y_{j,t+s}\frac{p_{j,t+s}}{p_{t+s}} - \varepsilon_{p,t+s} mc_{t+s} y_{j,t+s} - C^p_{j,t+s} \right) \right\}$$

Firm entry Bergin and Corsetti (2025a) model firm entry as follow. There is a fixed cost at entering the market:

$$K_t = \left(\frac{ne_t}{ne_{t-1}}\right)^\lambda \overline{K}$$

And the number of firm is, with δ the exogenous death shock:

$$n_{t+1} = (1 - \delta)(n_t + ne_t)$$

And firms will enter up to the point:

$$\pi_{j,t}(x) \geq K_t$$

This means that we need to define n_{t+1} as the number of firms and not $l_t(x)$ as it is now.

5.3.1 Solve

Demand faced by each intermediate good producer:

$$y^d_{j,t}(f) = \gamma_y \left(\frac{P_{G,t}}{e_t \tilde{p}_{j,t}(f)}\right)^\eta G_t(h) + (1-\gamma_y^*) \left(\frac{P^*_{G,t}}{p_{j,t}(f)}\right)^\eta G_t(f)$$
$$y^d_{j,t}(h) = \gamma_y^* \left(\frac{P^*_{G,t}}{\frac{1+\tau_i}{e_t} p_{j,t}(h)}\right)^\eta G_t(f) + (1-\gamma_y) \left(\frac{P_{G,t}}{p_{j,t}(h)}\right)^\eta G_t(h)$$

Interpretation:

- *Total demand:* Global quantity = sum of domestic demand and export demand, each weighted by the size of the respective final good sector (G_t, G_t^*).
- *CBAM* enters via the export price $\tilde{p}_{j,t}(f)$.

For home firms
1st step: choose abatement and define marginal cost, for a home intermediate firm.
The total variable cost $TC_{j,t}(h)$ is:

$$TC_{j,t}(h) = w_t(h) \left(\frac{y_{j,t}(h)}{\Gamma_t(h)}\right)^{\frac{1}{\alpha}} + \theta_{1,t}(h)\mu_{j,t}(h)^{\theta_2} y_{j,t}(h) + \tau_{e,t}(h)\sigma_t(h)(1-\mu_{j,t}(h))y_{j,t}(h)$$

Optimal abatement minimizes costs:

$$\frac{\partial TC_{j,t}(h)}{\partial \mu_{j,t}(h)} = 0 \quad \implies \quad \theta_{1,t}(h)\theta_2\mu_{j,t}(h)^{\theta_2 - 1} = \tau_{e,t}(h)\sigma_t(h)$$

Marginal cost $mc_{j,t}(h)$ is the derivative of total cost with respect to output:

$$mc_{j,t}(h) = \frac{\partial TC_{j,t}(h)}{\partial y_{j,t}(h)} = \frac{1}{\alpha} w_t(h) \frac{1}{y_{j,t}(h)} \left(\frac{y_{j,t}(h)}{\Gamma_t(h)}\right)^{\frac{1}{\alpha}} + \theta_{1,t}(h)\mu_{j,t}(h)^{\theta_2} + \tau_{e,t}(h)\sigma_t(h)(1-\mu_{j,t}(h))$$

Interpretation:

- *Optimal abatement:* Marginal cost of abatement = carbon tax. The firm abates until the cost of increasing $\mu_{j,t}$ equals the tax avoid on the last unit of emissions.
- *Marginal cost:* $mc_{j,t}(h)$ = sum of production costs (wages), abatement costs, and carbon tax.

2nd stage: set the price to maximize profit.

$$\max_{p_{j,t}(h)} \quad \mathbb{E}_t \left\{ \sum_{s=0}^{\infty} \beta^s (1-\vartheta)^s \left(y_{j,t+s}(h) \frac{p_{j,t+s}(h)}{P_{G,t+s}} - mc_{j,t+s}(h) y_{j,t+s}(h) - C^p_{j,t+s}(h) \right) \right\}$$

subject to the demand:

$$y_{j,t}(h) = p_{j,t}(h)^{-\eta} \underbrace{\left[(1-\gamma_y) P^\eta_{G,t} G_t(h) + \gamma_y^* \left(P^*_{G,t} \frac{e_t}{1+\tau_i} \right)^\eta G_t(f) \right]}_{\equiv \Omega_t}$$

Taking the derivative with respect to the chosen price $p_{j,t}(h)$:

$$\begin{aligned}
\frac{\partial \pi_{j,t}(h)}{\partial p_{j,t}(h)} = \quad & (1-\eta) p_{j,t}(h)^{-\eta} \frac{\Omega_t}{P_{G,t}} \\
& + \eta mc_{j,t}(h) p_{j,t}(h)^{-\eta-1} \Omega_t \\
& - \kappa \left(\frac{p_{j,t}(h)}{p_{j,t-1}(h)} - \pi^\star \right) \left(\frac{1}{p_{j,t-1}(h)} \right) \frac{y_t(h)}{l_t(h)} \\
& + \beta(1-\vartheta) \mathbb{E}_t \left[\kappa \left(\frac{p_{j,t+1}(h)}{p_{j,t}(h)} - \pi^\star \right) \left(\frac{p_{j,t+1}(h)}{p_{j,t}(h)^2} \right) \frac{y_{t+1}(h)}{l_{t+1}(h)} \right] = 0
\end{aligned}$$

$$\begin{aligned}
\kappa \left(\frac{p_{j,t}(h)}{p_{j,t-1}(h)} - \pi^\star \right) \frac{p_{j,t}(h)}{p_{j,t-1}(h)} \frac{y_t(h)}{l_t(h)} = \quad & (1-\eta) y_{j,t}(h) \frac{p_{j,t}(h)}{P_{G,t}} + \eta mc_{j,t}(h) y_{j,t}(h) \\
& + \beta(1-\vartheta) \mathbb{E}_t \left[\kappa \left(\frac{p_{j,t+1}(h)}{p_{j,t}(h)} - \pi^\star \right) \frac{p_{j,t+1}(h)}{p_{j,t}(h)} \frac{y_{t+1}(h)}{l_{t+1}(h)} \right]
\end{aligned}$$

Interpretation:

- *Market power:* Under flexible prices ($\kappa = 0$), the firm sets its price as a markup over marginal cost determined by demand elasticity η.
- *Adjustments costs:* LHS is today's adjustment costs while RHS features the Discounted expected cost of adjusting tomorrow. The firm balances the cost of changing prices today against the cost of delaying the adjustment to the next period.

For foreign firms
The foreign firm chooses its price $p_{j,t}(f)$ and abatement $\mu_{j,t}(f)$ jointly to maximize the expected discounted sum of profits, subject to Rotemberg adjustment costs.

$$\begin{aligned}
\max_{\{p_{j,t}(f), \mu_{j,t}(f)\}} \mathbb{E}_t \sum_{s=0}^{\infty} \beta^s (1-\vartheta)^s \Bigg\{ & p_{j,t+s}(f) y_{j,t+s}(f) - w_{t+s}(f) \left(\frac{y_{j,t+s}(f)}{\Gamma_{t+s}(f)} \right)^{\frac{1}{\alpha}} \\
& - \theta_{1,t+s}(f) \mu_{j,t+s}(f)^{\theta_2} y_{j,t+s}(f) - \tau_{e,t+s}(f) \sigma_{t+s}(f) (1-\mu_{j,t+s}(f)) y_{j,t+s}(f) \\
& - \frac{\kappa}{2} \left(\frac{p_{j,t+s}(f)}{p_{j,t+s-1}(f)} - \pi^\star \right)^2 \frac{y_{t+s}(f)}{l_{t+s}(f)} \Bigg\}
\end{aligned}$$

where the total demand $y_{j,t}(f)$ is the sum of domestic and export demand:

$$\begin{aligned}
y_{j,t}(f) = & (1-\gamma_y^*) \left(\frac{p_{j,t}(f)}{P_{G,t}^*} \right)^{-\eta} G_t(f) \\
& + \gamma_y \left(\frac{(1+\tau_i) p_{j,t}(f) + \max\{\tau_{e,t}(h) - \tau_{e,t}(f), 0\} \sigma_t(f) (1-\mu_{j,t}(f))}{P_{G,t}/e_t} \right)^{-\eta} G_t(h)
\end{aligned}$$

Define:

$$\tilde{p}_{j,t}(f) \equiv (1+\tau_i)p_{j,t}(f) + \max\{\tau_{e,t}(h) - \tau_{e,t}(f), 0\}\sigma_t(f)(1-\mu_{j,t}(f)) \quad \text{(Effective export price)}$$
$$y^{\text{exp}}_{j,t}(f) \equiv \gamma_y \left(\frac{e_t\,\tilde{p}_{j,t}(f)}{P_{G,t}}\right)^{-\eta} G_t(h) \quad \text{(Exported quantity)}$$
$$\chi_{j,t} \equiv \frac{y^{exp}_{j,t}(f)}{y_{j,t}(f)} \quad \text{(Export share)}$$
$$mc_{j,t}(f) \equiv \frac{1}{\alpha} w_t(f) \left(\frac{1}{\Gamma_t(f)}\right)^{\frac{1}{\alpha}} y_{j,t}(f)^{\frac{1}{\alpha}-1} + \theta_{1,t}(f)\mu_{j,t}(f)^{\theta_2} + \tau_{e,t}(f)\sigma_t(f)(1-\mu_{j,t}(f)) \quad \text{(Marginal cost)}$$

The simplified first-order condition is:

$$\theta_{1,t}(f)\theta_2\mu_{j,t}(f)^{\theta_2-1} = \tau_{e,t}(f)\sigma_t(f) + \eta\chi_{j,t}\left(\frac{p_{j,t}(f) - mc_{j,t}(f)}{\tilde{p}_{j,t}(f)}\right)\max\{\tau_{e,t}(h) - \tau_{e,t}(f), 0\}\sigma_t(f)$$

Taking the derivative of the intertemporal profit function with respect to the chosen price $p_{j,t}(f)$ yields:

$$\begin{aligned}
0 = \quad & y_{j,t}(f) + [p_{j,t}(f) - mc_{j,t}(f)]\frac{\partial y_{j,t}(f)}{\partial p_{j,t}(f)} \\
& - \kappa\left(\frac{p_{j,t}(f)}{p_{j,t-1}(f)} - \pi^\star\right)\left(\frac{1}{p_{j,t-1}(f)}\right)\frac{y_t(f)}{l_t(f)} \\
& + \beta(1-\vartheta)\mathbb{E}_t\left[\kappa\left(\frac{p_{j,t+1}(f)}{p_{j,t}(f)} - \pi^\star\right)\left(\frac{p_{j,t+1}(f)}{p_{j,t}(f)^2}\right)\frac{y_{t+1}(f)}{l_{t+1}(f)}\right]
\end{aligned}$$

where the partial derivative of demand is:

$$\begin{aligned}
\frac{\partial y_{j,t}(f)}{\partial p_{j,t}(f)} = & -\eta(1-\gamma_y^*)\left(\frac{p_{j,t}(f)}{P^*_{G,t}}\right)^{-\eta-1}\left(\frac{1}{P^*_{G,t}}\right)G_t(f) \\
& - \eta\gamma_y\left(\frac{\tilde{p}_{j,t}(f)}{P_{G,t}/e_t}\right)^{-\eta-1}\left(\frac{1+\tau_i}{P_{G,t}/e_t}\right)G_t(h)
\end{aligned}$$

Interpretation:

- *Abatement:* Marginal cost of reducing emissions = Domestic carbon tax savings + CBAM savings on exports. The incentive to abate is function of the export share $\chi_{j,t}$ of the firm.
- *Price Setting:* Analogous to the Home country, the firm balances the cost of adjusting today against the discounted expected cost of adjusting tomorrow (Rotemberg smoothing). The optimal price $p_{j,t}(f)$ is a function of the weighted sensitivities of Home and Foreign demands; the firm internalizes that CBAM creates a wedge in the demand slopes (price sensitivities), making the arbitrage dependent on the relative share of each market in total demand.

5.4 Public Sector

1. Home Country

The home government budget must be balanced each period. The government issues bonds, collects

domestic carbon taxes, and gathers revenues from the CBAM applied to foreign imports. All revenues are then redistributed as lump-sum transfers to high-productivity workers.
The home government budget constraint is:

$$\begin{aligned}&\int_0^{l_t(h)} b_{i,t}(h)di + \tau_{e,t}(h)\int_0^{l_t(h)} e_{j,t}(h)dj\\&+ e_t\int_0^{l_t(f)} \max\{\tau_{e,t}(h)-\tau_{e,t}(f),0\}\,\sigma_t(f)(1-\mu_{j,t}(f))y_{j,t}(f)dj\\&= \frac{r_{t-1}(h)}{\pi_t}\int_0^{l_t(h)} b_{i,t-1}(h)di + \int_0^{l_t(h)} T^e_{i,t}di\end{aligned}$$

By definition, redistributive policy is budget neutral from lump-sum tax on high-productivity to lump-sum transfers to low-productivity households.

$$T^S_{i,t}(h) = \frac{\omega}{1-\omega}d_{i,t}(h)$$

The central bank follows a standard Taylor rule to set the nominal interest rate:

$$\frac{r_t(h)}{r(h)} = \left(\frac{r_{t-1}(h)}{r(h)}\right)^{\rho}\left[\frac{\pi^\star(h)}{\pi(h)}\left(\frac{\pi_t(h)}{\pi^\star_t(h)}\right)^{\phi_\pi}\left(\frac{y_t(h)}{y^\star_t(h)}\right)^{\phi_y}\right]^{1-\rho}$$

with the variables:

- $\pi^\star$, gross inflation target. Deterministic process: $\pi^\star = \delta_{\pi^\star}\pi + (1-\delta_{\pi^\star})\pi^\star_{t-1}$
- $y^\star_t$, natural output, the one which would prevail with imperefect competition but flexible prices.

2. Foreign Country
The foreign government faces a symmetric budget constraint, but without CBAM revenues. It issues bonds and collects domestic carbon taxes, redistributing them to its employed workers.
The foreign government budget constraint is:

$$\begin{aligned}&\int_0^{l_t(f)} b^*_{i,t}(f)di + \tau_{e,t}(f)\int_0^{l_t(f)} e_{j,t}(f)dj\\&= \frac{r^*_{t-1}(f)}{\pi^*_t}\int_0^{l_t(f)} b^*_{i,t-1}(f)di + \int_0^{l_t(f)} T^{e*}_{i,t}di\end{aligned}$$

Symmetrically:

$$T^S_{i,t}(f) = \frac{\omega}{1-\omega}d_{i,t}(f)$$

The foreign central bank follows an analogous Taylor rule:

$$\frac{r^*_t(f)}{r^*(f)} = \left(\frac{r^*_{t-1}(f)}{r^*(f)}\right)^{\rho}\left[\frac{\pi^\star(f)}{\pi^*(f)}\left(\frac{\pi^*_t(f)}{\pi^\star_t(f)}\right)^{\phi_\pi}\left(\frac{y^*_t(f)}{y^\star_t(f)}\right)^{\phi_y}\right]^{1-\rho}$$

5.5 Aggregation

Home country
Aggregate consumption is defined as:

$$\int_0^{l_t(h)} c_{i,t} di = \int_0^{l_t(h)} \mathbb{P}[\theta_{i,t} = 0] c_{i,L,t} di + \int_0^{l_t(h)} \mathbb{P}[\theta_{i,t} = 1] c_{i,H,t} di$$

And because all consumers are identical, given their type:

$$c_t = \omega c_{L,t} + (1-\omega) c_{H,t}$$

Bonds are in zero net supply:

$$\int_0^{l_t(h)} b_{i,t}(h) di + \int_0^{l_t(f)} b_{i,t}(f) di = 0$$

Larbor market is such:

$$\int_0^{l_t(h)} \mathbb{P}[\theta_{i,t} = 1] n_{i,t} di = \int_0^{l_t(h)} n^d_{j,t} dj$$

and because all consumers are identical and all intermediates firms face the same prices and wage:

$$(1-\omega) n_{i,t} = n^d_t$$

Foreign country
Symetrically:

$$\int_0^{l_t(f)} c^*_{i,t} di = \int_0^{l_t(f)} \mathbb{P}[\theta^*_{i,t} = 0] c^*_{i,L,t} di + \int_0^{l_t(f)} \mathbb{P}[\theta^*_{i,t} = 1] c^*_{i,H,t} di$$

$$c^*_t = \omega c^*_{L,t} + (1-\omega) c^*_{H,t}$$

$$\int_0^{l_t(h)} b_{i,t}(f) di + \int_0^{l_t(f)} b^*_{i,t}(f) di = 0$$

$$\int_0^{l_t(f)} \mathbb{P}[\theta^*_{i,t} = 1] n^*_{i,t} di = \int_0^{l_t(f)} n^{d*}_{j,t} dj$$

$$(1-\omega) n^*_{i,t} = n^{d*}_t$$

Resource Constraints (Goods Market Clearing)
Production in a country is consumed in both countries, used to pay price adjustments costs, abatement costs and to hold foreign bonds while a fraction of firms profit is lost due to exit of the market.
Home Country:

$$\begin{aligned} y_t(h) = {} & l_t(h) c_t(h) + l_t(f) c^*_t(h) \\ & + \frac{\kappa}{2}(\pi_t(h) - \pi^\star(h))^2 y_t(h) + \theta_{1,t}(h) \mu_t(h)^{\theta_2} y_t(h) \\ & + \vartheta \Pi_t(h) + l_t(h) AC(b_t(f)) \end{aligned}$$

Foreign Country:

$$\begin{aligned} y_t(f) = & \ l_t(f)c_t^*(f) + l_t(h)c_t(f) \\ & + \frac{\kappa}{2}(\pi_t^*(f) - \pi^\star(f))^2 y_t(f) + \theta_{1,t}(f)\mu_t(f)^{\theta_2} y_t(f) \\ & + \vartheta\Pi_t^*(f) + l_t(f)AC(b_t^*(h)) \end{aligned}$$

5.6 Net Foreign Assets and Trade Balance

Home NFA Integrating over the budget constraints of all households from the home country, the aggregate household constraint is:

$$\begin{aligned} & l_t(h)c_t + \int_0^{l_t(h)} b_{i,t}(h)di + e_t \int_0^{l_t(h)} b_{i,t(f)}di \\ & \quad = \frac{r_{t-1}(h)}{\pi_t} \int_0^{l_t(h)} b_{i,t-1}(h)di + e_t \frac{r_{t-1}(f)}{\pi_t} \int_0^{l_t(h)} b_{i,t-1}(f)di \\ & \quad + l_t(h)AC(b_{i,t}(f)) + w_t n_t + (1-\vartheta)\Pi_t + l_t(h)T_t^e \end{aligned}$$

Substituting the government budget and the aggregate intermediate profit (Π_t), the domestic terms ($w_t n_t$, $\tau_{e,t}(h)e_t(h)$, and domestic bonds) cancel out. The equation becomes:

$$\begin{aligned} & l_t(h)c_t + e_t l_t(h)b_t(f) \\ & \quad = e_t \frac{r_{t-1}(f)}{\pi_t} l_t(h)b_{t-1}(f) + l_t(h)AC(b_{i,t}(f)) + l_t(h)p_{j,t}(h)y_{j,t}^s(h) \\ & \quad - p_{j,t}(h)\left[l_t(h)\theta_{1,t}(h)\mu_{j,t,h}^{\theta_2(h)} y_{j,t}^s(h) + \frac{\kappa}{2}\left(\frac{p_{j,t}(h)}{p_{j,t-1}(h)} - \pi^\star\right)^2 l_t(h)y_{j,t}^s(h)\right] \\ & \quad - \vartheta\Pi_t + e_t \max\{\tau_{e,t}(h) - \tau_{e,t}(f), 0\}\sigma_t(f)(1-\mu_{j,t}(f))l_t(h)y_{j,t}^{exp}(f) \end{aligned}$$

Intermediate output from the representative firm j is sold to the home and foreign final sectors:

$$l_t(h)y_{j,t}^s(h) = l_t(h)y_{j,t}^d(h) + l_t(f)y_{j,t}^{d*}(h)$$

Substituting this into the consolidated budget:

$$\begin{aligned} & l_t(h)c_t + e_t l_t(h)b_t(f) \\ & \quad = e_t \frac{r_{t-1}(f)}{\pi_t} l_t(h)b_{t-1}(f) + l_t(h)AC(b_{i,t}(f)) + p_{j,t}(h)[l_t(h)y_{j,t}^d(h) + l_t(f)y_{j,t}^{d*}(h)] \\ & \quad - p_{j,t}(h)\left[l_t(h)\theta_{1,t}(h)\mu_{j,t,h}^{\theta_2(h)} y_{j,t}^s(h) + \frac{\kappa}{2}\left(\frac{p_{j,t}(h)}{p_{j,t-1}(h)} - \pi^\star\right)^2 l_t(h)y_{j,t}^s(h)\right] \\ & \quad - \vartheta\Pi_t + e_t \max\{\tau_{e,t}(h) - \tau_{e,t}(f), 0\}\sigma_t(f)(1-\mu_{j,t}(f))l_t(h)y_{j,t}^{exp}(f) \end{aligned}$$

The zero-profit condition for the Home final good sector is:

$$\begin{aligned} p_t(h)y_t(h) = & \ p_{j,t}(h)l_t(h)y_{j,t}^d(h) + e_t(1+\tau_i)p_{j,t}(f)l_t(h)y_{j,t}^d(f) \\ & + e_t \max\{\tau_{e,t}(h) - \tau_{e,t}(f), 0\}\sigma_t(f)(1-\mu_{j,t}(f))l_t(h)y_{j,t}^{exp}(f) \end{aligned}$$

Substituting the term $[p_{j,t}(h)l_t(h)y^d_{j,t}(h) + \text{CBAM income}]$:

$$\begin{aligned}
&l_t(h)c_t + e_t l_t(h)b_t(f) \\
&\quad = e_t \frac{r_{t-1}(f)}{\pi_t} l_t(h)b_{t-1}(f) + l_t(h)AC(b_{i,t}(f)) + p_t(h)y_t(h) \\
&\quad - e_t(1+\tau_i)p_{j,t}(f)l_t(h)y^d_{j,t}(f) + p_{j,t}(h)l_t(f)y^{d*}_{j,t}(h) \\
&\quad - p_{j,t}(h)\left[l_t(h)\theta_{1,t}(h)\mu^{\theta_2(h)}_{j,t,h}y^s_{j,t}(h) + \frac{\kappa}{2}\left(\frac{p_{j,t}(h)}{p_{j,t-1}(h)} - \pi^\star\right)^2 l_t(h)y^s_{j,t}(h)\right] - \vartheta\Pi_t
\end{aligned}$$

Recall the home resource constraint:

$$\begin{aligned}
p_t(h)y_t(h) &= p_t(h)l_t(h)c_t(h) + p_t(h)l_t(f)c^*_t(h) + \vartheta\Pi_t + l_t(h)AC(b_{i,t}(f)) \\
&\quad + p_{j,t}(h)\left[l_t(h)\theta_{1,t}(h)\mu^{\theta_2(h)}_{j,t,h}y^s_{j,t}(h) + \frac{\kappa}{2}\left(\frac{p_{j,t}(h)}{p_{j,t-1}(h)} - \pi^\star\right)^2 l_t(h)y^s_{j,t}(h)\right]
\end{aligned}$$

Plugging this into the budget equation, abatement, Rotemberg, and foreign bond holding costs cancel out. The same applies to $\vartheta\Pi_t$. Decomposing $l_t(h)c_t = p_t(h)l_t(h)c_t(h) + e_t p^*_t(f)l_t(h)c_t(f)$ and canceling the domestic consumption $p_t(h)l_t(h)c_t(h)$, we obtain:

$$\begin{aligned}
e_t p^*_t(f)l_t(h)c_t(f) + e_t l_t(h)b_t(f) &= e_t \frac{r_{t-1}(f)}{\pi_t} l_t(h)b_{t-1}(f) \\
&\quad + p_t(h)l_t(f)c^*_t(h) + p_{j,t}(h)l_t(f)y^{d*}_{j,t}(h) - e_t(1+\tau_i)p_{j,t}(f)l_t(h)y^d_{j,t}(f)
\end{aligned}$$

Isolating the NFA flow and the Trade Balances for final and intermediate goods:

$$e_t l_t(h)b_t(f) - e_t \frac{r_{t-1}(f)}{\pi_t} l_t(h)b_{t-1}(f) = TB^{final}_t + TB^{inter}_t$$

where:

$$\begin{aligned}
TB^{final}_t &\equiv p_t(h)l_t(f)c^*_t(h) - e_t p^*_t(f)l_t(h)c_t(f) \\
TB^{inter}_t &\equiv p_{j,t}(h)l_t(f)y^{d*}_{j,t}(h) - e_t(1+\tau_i)p_{j,t}(f)l_t(h)y^d_{j,t}(f)
\end{aligned}$$

Foreign NFA
By symmetry, we aggregate the Foreign household budget, government budget, and firm profits.

$$\begin{aligned}
&l_t(f)c^*_t + \frac{1}{e_t} l_t(f)b^*_{i,t}(h) \\
&\quad = \frac{1}{e_t}\frac{r_{t-1}(h)}{\pi^*_t} l_t(f)b^*_{i,t-1}(h) + l_t(f)p^*_{j,t}(f)y^{s*}_{j,t}(f) \\
&\quad - p^*_{j,t}(f)\left[l_t(f)\theta^*_{1,t}(f)\mu^{\theta^*_2(f)}_{j,t,f}y^{s*}_{j,t}(f) + \frac{\kappa}{2}\left(\frac{p^*_{j,t}(f)}{p^*_{j,t-1}(f)} - \pi^{\star*}\right)^2 l_t(f)y^{s*}_{j,t}(f)\right] \\
&\quad - \vartheta\Pi^*_t
\end{aligned}$$

Using the Foreign resource constraint and the zero-profit condition for the Foreign final good sector, we decompose Foreign consumption into domestic and imported components: $l_t(f)c^*_t = p^*_t(f)l_t(f)c^*_t(f) + \frac{p_t(h)}{e_t}l_t(f)c^*_t(h)$. Canceling domestic terms yields:

$$\frac{1}{e_t} l_t(f)b^*_{i,t}(h) - \frac{1}{e_t}\frac{r_{t-1}(h)}{\pi^*_t} l_t(f)b^*_{i,t-1}(h) = TB^{*,final}_t + TB^{*,inter}_t$$

where the Foreign trade balances are:

$$TB_t^{*,final} \equiv p_t^*(f) l_t(h) c_t(f) - \frac{p_t(h)}{e_t} l_t(f) c_t^*(h)$$
$$TB_t^{*,inter} \equiv p_{j,t}^*(f) l_t(h) y_{j,t}^d(f) - \frac{(1+\tau_i) p_{j,t}(h)}{e_t} l_t(f) y_{j,t}^{d*}(h)$$

The evolution of Foreign Net Foreign Assets is:

$$NFA_t^* = \frac{e_{t-1}}{e_t} \frac{r_{t-1}(h)}{\pi_t^*} NFA_{t-1}^* + TB_t^{*,final} + TB_t^{*,inter}$$

5.7 Walras law

Verified from the global bond market clearing condition:

$$NFA_t + e_t NFA_t^* = 0$$

Summary

5.7.1 Variables

Block	Variables	#
1. Households	Home consumption by type and good: $c_{i,H,t}(h),\ c_{i,H,t}(f),\ c_{i,L,t}(h),\ c_{i,L,t}(f)$	4
	Home labor supply: $n_{i,t}$	1
	Home bond holdings: $b_{i,t}(h),\ b_{i,t}(f)$	2
	Foreign consumption by type and good: $c^*_{i,H,t}(h),\ c^*_{i,H,t}(f),\ c^*_{i,L,t}(h),\ c^*_{i,L,t}(f)$	4
	Foreign labor supply: $n^*_{i,t}$	1
	Foreign bond holdings: $b^*_{i,t}(h),\ b^*_{i,t}(f)$	2
	Subtotal	**14**
2. Final Good Firms	Home output and intermediate demand: $y_t(h),\ G_t(h)$	2
	Home demand for each intermediate: $y^d_{j,t}(h),\ y^d_{j,t}(f)$	2
	Foreign output and intermediate demand: $y_t(f),\ G_t(f)$	2
	Foreign demand for each intermediate: $y^{d*}_{j,t}(h),\ y^{d*}_{j,t}(f)$	2
	Subtotal	**8**
3. Intermediate Firms	Home and foreign supply: $y^s_{j,t}(h),\ y^s_{j,t}(f)$	2
	Labor demand: $n^d_{j,t}(h),\ n^d_{j,t}(f)$	2
	Abatement: $\mu_{j,t}(h),\ \mu_{j,t}(f)$	2
	Prices: $p_{j,t}(h),\ p_{j,t}(f)$	2
	Subtotal	**8**
4. Policy & Macro	Wages: $w_t,\ w^*_t$	2
	Nominal interest rates: $r_t(h),\ r^*_t(f)$	2
	Government transfers: $T^e_t,\ T^{e*}_t$	2
	Exchange rate : e_t	1
	Subtotal	**7**
Total decision variables		**37**

5.7.2 Definition Variables and Equations

Block	**Variables**	#
Households	Country aggregates: $c_t,\ c_t^*$	2
	Bilateral final good demands: $c_t(h),\ c_t(f),\ c_t^*(h),\ c_t^*(f)$	4
	Subtotal	**6**
Price Indices	Consumer price indices: $P_t,\ P_t^*$	2
	CPI inflation: $\pi_t,\ \pi_t^*$	2
	Intermediate price indices: $P_{G,t},\ P_{G,t}^*$	2
	Subtotal	**6**
Intermediate Firms	Emissions: $e_{j,t}(h),\ e_{j,t}(f)$	2
	Marginal costs: $mc_{j,t}(h),\ mc_{j,t}(f)$	2
	Effective export price, export share, demand derivative: $\tilde{p}_{j,t}(f),\ \chi_{j,t},\ \Lambda_{j,t}(f)$	3
	PPI inflation: $\pi_{j,t}(h),\ \pi_{j,t}^*(f)$	2
	Subtotal	**9**
Government	Aggregate profits: $\Pi_t,\ \Pi_t^*$	2
	Subtotal	**2**
Total definition variables		**23**

5.7.3 Optimality Conditions

Block 1 — Households

1. **Home Euler (type H):**

$$c_{i,H,t}^{-\sigma} = \mathbb{E}_t \left\{ \beta \frac{r_t(h)}{\pi_{t+1}} \left[(1-\omega)\, c_{i,H,t+1}^{-\sigma} + \omega\, c_{i,L,t+1}^{-\sigma} \right] \right\}$$

2. **Home Euler (type L):**

$$c_{i,L,t}^{-\sigma} = \mathbb{E}_t \left\{ \beta \frac{r_t(h)}{\pi_{t+1}} \left[h\, c_{i,H,t+1}^{-\sigma} + (1-h)\, c_{i,L,t+1}^{-\sigma} \right] \right\}$$

3. **Home labor supply:**

$$\psi_t\, n_{i,t}^{\sigma_n} = c_{i,H,t}^{-\sigma}\, w_t$$

4. **Home demand for bonds:**

$$\frac{r_t(h)}{\pi_{t+1}} = e_{t+1}\, \frac{r_t(f)}{\pi_{t+1}} + \frac{\partial AC(b_{i,t}(f))}{\partial b_{i,t}(f)}$$

5. **Home demand for domestic final good:**

$$c_{i,t}(h) = \left(\frac{P_t}{p_t(h)}\right)^{\phi} c_{i,t}$$

6. **Home demand for imported final good:**

$$c_{i,t}(f) = \left(\frac{P_t}{e_t\, p_t^*(f)}\right)^{\phi} c_{i,t}$$

7. **Foreign Euler (type H):**

$$(c_{i,H,t}^*)^{-\sigma} = \mathbb{E}_t \left\{\beta \frac{r_t^*(f)}{\pi_{t+1}^*} \left[(1-\omega)\,(c_{i,H,t+1}^*)^{-\sigma} + \omega\,(c_{i,L,t+1}^*)^{-\sigma}\right]\right\}$$

8. **Foreign Euler (type L):**

$$(c_{i,L,t}^*)^{-\sigma} = \mathbb{E}_t \left\{\beta \frac{r_t^*(f)}{\pi_{t+1}^*} \left[h\,(c_{i,H,t+1}^*)^{-\sigma} + (1-h)\,(c_{i,L,t+1}^*)^{-\sigma}\right]\right\}$$

9. **Foreign labor supply:**

$$\psi_t^*\,(n_{i,t}^*)^{\sigma_n} = (c_{i,H,t}^*)^{-\sigma}\, w_t^*$$

10. **Foreign demand for bonds:**

$$\frac{r_t^*(f)}{\pi_{t+1}^*} = \frac{1}{e_{t+1}}\,\frac{r_t(h)}{\pi_{t+1}^*} + \frac{\partial AC(b_{i,t}^*(h))}{\partial b_{i,t}^*(h)}$$

11. **Foreign demand for foreign final good:**

$$c_{i,t}^*(f) = \left(\frac{P_t^*}{p_t^*(f)}\right)^{\phi} c_{i,t}^*$$

12. **Foreign demand for imported final good:**

$$c_{i,t}^*(h) = \left(\frac{P_t^*}{p_t(h)/e_t}\right)^{\phi} c_{i,t}^*$$

Block 2 — Final Good Firms

13. **Home demand for domestic intermediates:**

$$y_{j,t}^d(h) = \left(\frac{p_{j,t}(h)}{P_{G,t}}\right)^{-\eta} G_t(h)$$

14. **Home demand for foreign intermediates:**

$$y_{j,t}^d(f) = \left(\frac{e_t\,\tilde{p}_{j,t}(f)}{P_{G,t}}\right)^{-\eta} G_t(h)$$

15. **Home final good firm FOC ($G_t(h)$):**

$$G_t(h) = \left(\frac{\zeta\, p_t(h)\, \alpha_h\, l_t(h)^{1-\zeta}}{P_{G,t}} \right)^{\frac{1}{1-\zeta}}$$

16. **Home final good output:**

$$y_t(h) = \alpha_h\, G_t(h)^{\zeta}\, l_t(h)^{1-\zeta}$$

17. **Foreign demand for foreign intermediates:**

$$y^{d*}_{j,t}(f) = \left(\frac{p_{j,t}(f)}{P^*_{G,t}} \right)^{-\eta} G_t(f)$$

18. **Foreign demand for home intermediates:**

$$y^{d*}_{j,t}(h) = \left(\frac{p_{j,t}(h)/e_t}{P^*_{G,t}} \right)^{-\eta} G_t(f)$$

19. **Foreign final good firm FOC ($G_t(f)$):**

$$G_t(f) = \left(\frac{\zeta\, p^*_t(f)\, \alpha_f\, l_t(f)^{1-\zeta}}{P^*_{G,t}} \right)^{\frac{1}{1-\zeta}}$$

20. **Foreign final good output:**

$$y_t(f) = \alpha_f\, G_t(f)^{\zeta}\, l_t(f)^{1-\zeta}$$

Block 3 — Intermediate Firms

21. **Home labor demand:**

$$n^d_{j,t}(h) = \left(\frac{y^s_{j,t}(h)}{\Gamma_t(h)\, l_t(h)^{1-\alpha}} \right)^{1/\alpha}$$

22. **Foreign labor demand:**

$$n^d_{j,t}(f) = \left(\frac{y^s_{j,t}(f)}{\Gamma_t(f)\, l_t(f)^{1-\alpha}} \right)^{1/\alpha}$$

23. **Home optimal abatement:**

$$\theta_{1,t}(h)\, \theta_2\, \mu_{j,t}(h)^{\theta_2 - 1} = \tau_{e,t}(h)\, \sigma_t(h)$$

24. **Foreign optimal abatement:**

$$\theta_{1,t}(f)\, \theta_2\, \mu_{j,t}(f)^{\theta_2 - 1} = \tau_{e,t}(f)\, \sigma_t(f) + \eta\, \chi_{j,t} \left(\frac{p_{j,t}(f) - mc_{j,t}(f)}{\tilde{p}_{j,t}(f)} \right) \max\{\tau_{e,t}(h) - \tau_{e,t}(f), 0\}\, \sigma_t(f)$$

25. **Home intermediate firm pricing:**

$$\kappa\left(\pi_{j,t}(h)-\pi^{\star}(h)\right)\pi_{j,t}(h)\frac{y^s_{j,t}(h)}{l_t(h)} = (1-\eta)\,y^s_{j,t}(h)\frac{p_{j,t}(h)}{P_{G,t}} + \eta\, mc_{j,t}(h)\, y^s_{j,t}(h)$$
$$+\,\beta(1-\vartheta)\,\mathbb{E}_t\left[\kappa\left(\pi_{j,t+1}(h)-\pi^{\star}(h)\right)\pi_{j,t+1}(h)\frac{y^s_{j,t+1}(h)}{l_{t+1}(h)}\right]$$

26. **Foreign intermediate firm pricing:**

$$0 = y^s_{j,t}(f) + \left[p_{j,t}(f)-mc_{j,t}(f)\right]\Lambda_{j,t}(f) - \kappa\left(\pi^*_{j,t}(f)-\pi^{\star}(f)\right)\pi^*_{j,t}(f)\frac{y^s_{j,t}(f)}{l_t(f)}$$
$$+\,\beta(1-\vartheta)\,\mathbb{E}_t\left[\kappa\left(\pi^*_{j,t+1}(f)-\pi^{\star}(f)\right)\frac{p_{j,t+1}(f)}{p_{j,t}(f)^2}\frac{y^s_{j,t+1}(f)}{l_{t+1}(f)}\right]$$

5.7.4 Market Clearing & Policy

27. **Home intermediate goods market clearing:**

$$y^s_{j,t}(h) = y^d_{j,t}(h) + y^{d*}_{j,t}(h)$$

28. **Foreign intermediate goods market clearing:**

$$y^s_{j,t}(f) = y^d_{j,t}(f) + y^{d*}_{j,t}(f)$$

29. **Home final good market clearing:**

$$y_t(h) = l_t(h)\,c_t(h) + l_t(f)\,c^*_t(h) + \frac{\kappa}{2}(\pi_{j,t}(h)-\pi^{\star}(h))^2 y^s_{j,t}(h)$$
$$+\,\theta_{1,t}(h)\,\mu_{j,t}(h)^{\theta_2}\,y^s_{j,t}(h) + \vartheta\,\Pi_t + l_t(h)\,AC(b_{i,t}(f))$$

30. **Foreign final good market clearing:**

$$y_t(f) = l_t(f)\,c^*_t(f) + l_t(h)\,c_t(f) + \frac{\kappa}{2}(\pi^*_{j,t}(f)-\pi^{\star}(f))^2 y^s_{j,t}(f)$$
$$+\,\theta_{1,t}(f)\,\mu_{j,t}(f)^{\theta_2}\,y^s_{j,t}(f) + \vartheta\,\Pi^*_t + l_t(f)\,AC(b^*_{i,t}(h))$$

31. **Home labor market clearing:**

$$(1-\omega)\,n_{i,t} = n^d_{j,t}(h)$$

32. **Foreign labor market clearing:**

$$(1-\omega)\,n^*_{i,t} = n^d_{j,t}(f)$$

33. **Home bond market clearing:**

$$\int_0^{l_t(h)} b_{i,t}(h)\,di + \int_0^{l_t(f)} b_{i,t}(f)\,di = 0$$

34. **Foreign bond market clearing:**

$$\int_0^{l_t(f)} b^*_{i,t}(f)\,di + \int_0^{l_t(h)} b_{i,t}(f)\,di = 0$$

35. **Home Taylor rule:**

$$\frac{r_t(h)}{r(h)} = \left(\frac{r_{t-1}(h)}{r(h)}\right)^{\rho} \left[\frac{\pi^\star(h)}{\pi_t(h)} \left(\frac{\pi_t(h)}{\pi_t^\star(h)}\right)^{\phi_\pi} \left(\frac{y_t(h)}{y_t^\star(h)}\right)^{\phi_y}\right]^{1-\rho}$$

36. **Foreign Taylor rule:**

$$\frac{r^*_t(f)}{r^*(f)} = \left(\frac{r^*_{t-1}(f)}{r^*(f)}\right)^{\rho} \left[\frac{\pi^\star(f)}{\pi^*_t(f)} \left(\frac{\pi^*_t(f)}{\pi_t^\star(f)}\right)^{\phi_\pi} \left(\frac{y^*_t(f)}{y_t^\star(f)}\right)^{\phi_y}\right]^{1-\rho}$$

37. **Home government budget:**

$$\begin{aligned}
&\int_0^{l_t(h)} b_{i,t}(h)\,di + \tau_{e,t}(h) \int_0^{l_t(h)} e_{j,t}(h)\,dj \\
&+ e_t \int_0^{l_t(f)} \max\{\tau_{e,t}(h) - \tau_{e,t}(f), 0\}\, \sigma_t(f)\,(1-\mu_{j,t}(f))\, y^s_{j,t}(f)\,dj \\
&= \frac{r_{t-1}(h)}{\pi_t} \int_0^{l_t(h)} b_{i,t-1}(h)\,di + \int_0^{l_t(h)} \mathbb{P}[\theta_{i,t}=1]\, T^e_{i,t}\,di
\end{aligned}$$

38. **Foreign government budget:**

$$\begin{aligned}
&\int_0^{l_t(f)} b^*_{i,t}(f)\,di + \tau_{e,t}(f) \int_0^{l_t(f)} e_{j,t}(f)\,dj \\
&= \frac{r^*_{t-1}(f)}{\pi^*_t} \int_0^{l_t(f)} b^*_{i,t-1}(f)\,di + \int_0^{l_t(f)} \mathbb{P}[\theta^*_{i,t}=1]\, T^{e*}_{i,t}\,di
\end{aligned}$$

5.7.5 Definition equations

37. **Home aggregate consumption:**

$$c_t = \omega\, c_{L,t} + (1-\omega)\, c_{H,t}$$

38. **Foreign aggregate consumption:**

$$c^*_t = \omega\, c^*_{L,t} + (1-\omega)\, c^*_{H,t}$$

39. **Home demand for domestic final good (aggregate):**

$$c_t(h) = \left(\frac{P_t}{p_t(h)}\right)^{\phi} c_t$$

40. **Home demand for imported final good (aggregate):**

$$c_t(f) = \left(\frac{P_t}{e_t\, p_t^*(f)}\right)^{\phi} c_t$$

41. **Foreign demand for foreign final good (aggregate):**

$$c_t^*(f) = \left(\frac{P_t^*}{p_t^*(f)}\right)^{\phi} c_t^*$$

42. **Foreign demand for imported final good (aggregate):**

$$c_t^*(h) = \left(\frac{P_t^*}{p_t(h)/e_t}\right)^{\phi} c_t^*$$

43. **Home CPI:**

$$P_t = \left[p_t(h)^{1-\phi} + (e_t\, p_t^*(f))^{1-\phi}\right]^{\frac{1}{1-\phi}}$$

44. **Foreign CPI:**

$$P_t^* = \left[p_t^*(f)^{1-\phi} + \left(\frac{p_t(h)}{e_t}\right)^{1-\phi}\right]^{\frac{1}{1-\phi}}$$

45. **Home CPI inflation:**

$$\pi_t = \frac{P_t}{P_{t-1}}$$

46. **Foreign CPI inflation:**

$$\pi_t^* = \frac{P_t^*}{P_{t-1}^*}$$

47. **Home intermediate aggregator price index:**

$$P_{G,t} = \left(p_{j,t}(h)^{1-\eta} + (e_t\, \tilde{p}_{j,t}(f))^{1-\eta}\right)^{\frac{1}{1-\eta}}$$

48. **Foreign intermediate aggregator price index:**

$$P_{G,t}^* = \left(p_{j,t}(f)^{1-\eta} + \left(\frac{p_{j,t}(h)}{e_t}\right)^{1-\eta}\right)^{\frac{1}{1-\eta}}$$

49. **Home firm emissions:**

$$e_{j,t}(h) = \sigma_t(h)\,(1-\mu_{j,t}(h))\, y_{j,t}^s(h)$$

50. **Foreign firm emissions:**

$$e_{j,t}(f) = \sigma_t(f)\,(1-\mu_{j,t}(f))\, y_{j,t}^s(f)$$

51. **Home marginal cost:**

$$mc_{j,t}(h) = \frac{w_t}{\alpha}\,\Gamma_t(h)^{-1/\alpha}\, y_{j,t}^s(h)^{\frac{1}{\alpha}-1} + \theta_{1,t}(h)\,\mu_{j,t}(h)^{\theta_2} + \tau_{e,t}(h)\,\sigma_t(h)\,(1-\mu_{j,t}(h))$$

52. **Foreign marginal cost:**

$$mc_{j,t}(f) = \frac{w_t^*}{\alpha}\,\Gamma_t(f)^{-1/\alpha}\,y_{j,t}^s(f)^{\frac{1}{\alpha}-1} + \theta_{1,t}(f)\,\mu_{j,t}(f)^{\theta_2} + \tau_{e,t}(f)\,\sigma_t(f)\,(1-\mu_{j,t}(f))$$

53. **Effective export price (CBAM):**

$$\tilde{p}_{j,t}(f) = (1+\tau_i)\,p_{j,t}(f) + \max\{\tau_{e,t}(h) - \tau_{e,t}(f), 0\}\,\sigma_t(f)\,(1-\mu_{j,t}(f))$$

54. **Export share:**

$$\chi_{j,t} = \frac{y_{j,t}^d(f)}{y_{j,t}^s(f)}$$

55. **Demand derivative:**

$$\Lambda_{j,t}(f) = -\eta\left(\frac{p_{j,t}(f)}{P_{G,t}^*}\right)^{-\eta-1}\frac{G_t(f)}{P_{G,t}^*} - \eta\left(\frac{\tilde{p}_{j,t}(f)}{P_{G,t}/e_t}\right)^{-\eta-1}\frac{1+\tau_i}{P_{G,t}/e_t}\,G_t(h)$$

56. **Home PPI inflation:**

$$\pi_{j,t}(h) = \frac{p_{j,t}(h)}{p_{j,t-1}(h)}$$

57. **Foreign PPI inflation:**

$$\pi_{j,t}^*(f) = \frac{p_{j,t}(f)}{p_{j,t-1}(f)}$$

58. **Home aggregate profits:**

$$\Pi_t = p_{j,t}(h)\,y_{j,t}^s(h) - w_t\,n_{j,t}^d(h) - \theta_{1,t}(h)\,\mu_{j,t}(h)^{\theta_2}\,y_{j,t}^s(h) - \frac{\kappa}{2}\,(\pi_{j,t}(h) - \pi^\star(h))^2\,\frac{y_{j,t}^s(h)}{l_t(h)}$$

59. **Foreign aggregate profits:**

$$\Pi_t^* = p_{j,t}(f)\,y_{j,t}^s(f) - w_t^*\,n_{j,t}^d(f) - \theta_{1,t}(f)\,\mu_{j,t}(f)^{\theta_2}\,y_{j,t}^s(f) - \frac{\kappa}{2}\left(\pi_{j,t}^*(f) - \pi^\star(f)\right)^2\frac{y_{j,t}^s(f)}{l_t(f)}$$

6 Exercises

- Effect of the introduction of CBAM on inflation.
- Non cooperative monetary policy ==> can foreign country offset the CBAM through devaluation?
- Monetary policy : comparison Taylor rule, Ramsey cooperative, maximization of national welfare given policy rule in the foreign country?
- Difference between CBAM and tariff?
- No CBAM, introduction of carbon tax, observe the production reallocation towards other country and model carbon leakage.

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